

UNIT V – NUCLEAR ENERGY SOURCE

Syllabus: General Properties of Nucleus (Size, Mass, Density, Charge) – Mass Defect – Binding Energy - Disintegration in fission –Nuclear Reactor: Materials Used in Nuclear Reactors. – PWR – BWR – FBTR. Nuclear fusion reactions for fusion reactors-D-D and D-T reactions, p-p and C-N cycle) Basic principles of Nuclear Fusion reactors.

Unit – V Nuclear Energy Source

Pondicherry University 2 Marks Question and Answers

Part - A

1. What are prompt neutrons? (May 2012)

In nuclear fission reactions, neutrons emitted instantaneously by a nucleus undergoing fission are called prompt neutrons. These prompt neutrons are released directly from fission within 10^{-13} seconds of the fission event

2. What is mass defect? (May 2013,2014,2015,2017,2018 Jan 2014, 2016,Nov 2017,2018,2020)

The difference between the actual mass of the nucleus and the sum of the masses of constituent nucleons is called mass defect.

3. What is the principle used in D-D nuclear reactor? (Jan 2014)

Thermonuclear reactions fusion reaction is the principle used in D-D nuclear reactor.

4. Determine the energy equivalent of 1 a.m.u. (Jan 2011,2015), (May 2011), (May 2012)

The energy equivalent of 1 amu is 931 MeV.

5. What do you mean by nuclear fusion reaction? (Jan 2011, May 2017,Nov2017) (Or) What is meant by nuclear fusion? (Nov 2018) (OR) Explain nuclear fusion reactions?(May 2015)

Nuclear fusion is a nuclear reaction in which two lighter nuclei fuse together to form a heavier and stable nucleus with the release of large amount of energy.



For example when the two deuterons are brought together to form single Helium (α -particle) the following reaction takes place with the liberation of 24 MeV energy.

6. What is binding energy? (OR) What is a nuclear binding energy?(2016 Nov, Dec 2019)

(Jan 2013,2014,2015, 2016,May2014,2015,2016,2017,2018,Nov 2017,2018,2020)

The amount of energy required to break up the atom into its constituents is equal to the mass defect the energy equivalent of mass defect is a measure of binding energy of the atom.

$$\text{B.E} = \Delta mc^2 = [Z m_p + N m_n - m] c^2$$

The binding energy is expressed in MeV using the relation.

$$1\text{amu}=931.478 \text{ MeV}$$

7. The process in which nucleus of a heavy atom breaks into two, more or less equal segments with release of energy is known as-(d) Fission (Jan 2013)

a) Fusion b) Ionization c) radioactivity d) fission

8. Define the term multiplication factor. (May 2011)

Multiplication factor is the ratio of number of neutrons produced in any particular generation to number of neutrons produced in the preceding generation

$$K = \frac{\text{number of neutrons produced in any particular generation}}{\text{number of neutrons produced in the preceding generation}}$$

When $k=1$, fission chain reaction will be critical or steady

When $k>1$ it will be “building up” or super critical.

When $k<1$ it will be dying down or sub critical.

9. What are the essential components of a nuclear reactor? (May 2013)(or) Name the materials used in nuclear reactor.(Jan 2014,2015)

1. Reactor core
2. Reflector
3. Moderator
4. Coolant
5. Control material
6. Shielding.

10. Explain the general properties of nucleus (Jan 2016) (Or) What are the properties of nucleus? (Nov 2018,May 2018,2014)

(i). Nuclear size

If the nucleus is assumed to be spherical, an empirical relation is found to hold good between the radius of the nucleus R and its mass number A . It is given by

$$R \propto A^{1/3}$$

$$R = r_0 A^{1/3} \quad \dots(1)$$

Where r_0 is the constant of proportionality and is equal to $1.3F$

(1 Fermi, $F = 10^{-15} \text{ m}$)

(ii). Nuclear density

The nuclear density ρ_N can be calculated from the mass and size of the nucleus.

$$\rho_N = \frac{\text{Nuclear mass}}{\text{Nuclear volume}}$$

Nuclear mass = Am_N where, A = mass number and m_N = mass of one nucleon and is approximately equal to 1.67×10^{-27} kg.

$$\begin{aligned} \text{Nuclear volume} &= \frac{4}{3} \pi R^3 \\ &= \frac{4}{3} \pi (r_0 A^{1/3})^3 \\ &= \frac{4}{3} \pi (r_0 A^{1/3})^3 \quad \dots(1) \\ \therefore \rho_N &= \frac{Am_N}{\frac{4}{3} \pi r_0^3 A} \end{aligned}$$

Substituting the known values, the nuclear density is calculated as 1.816×10^{17} kg m⁻³ which is almost a constant for all the nuclei irrespective of its size.

(iii). Nuclear charge

The charge of a nucleus is due to the protons present in it. Each proton has a positive charge equal to 1.6×10^{-19} C.

$\therefore$ The nuclear charge = Ze , where Z is the atomic number.

(iv). Nuclear mass

As the nucleus contains protons and neutrons, the mass of the nucleus is assumed to be the mass of its constituents. Assumed nuclear mass = $Zm_p + Nm_n$, where m_p and m_n are the mass of a proton and a neutron respectively. However, from the measurement of mass by mass spectrometers, it is found that the mass of a stable nucleus (m) is less than the total mass of the nucleons. i.e., mass of a nucleus, $m < (Zm_p + Nm_n)$, $Zm_p + Nm_n - m = \Delta m$, where Δm is the mass defect .

11. What are the required specifications for the reflector? Mention the example.(Jan 2016)

Reflector should have

1. Materials of high scattering cross- section.
2. Low absorption cross section.

Materials used:

1. Heavy water 2. Water 3. Graphite 4.Beryllium oxide.

12. Difference between nuclear fission and nuclear fusion. Give some examples? (Jan16,May17, 2018,2014,2015Nov 2018)

(OR) State the law of disintegration

S.No.	Nuclear fission	Nuclear fusion
1.	Huge amount of energy is liberated	Huge amount of energy is liberated
2.	A heavy nucleus is split up in to two nuclei.	Two lighter nuclei are fused together.
3.	This process is possible even at room temperature	This process is possible only at very high temperature
4.	The link of this process are neutrons	The link of this process are protons
5.	The energy per nucleon is 0.85 MeV	The energy per nucleon is 6.75MeV
6.	Example: ${}_{92}\text{U}^{235} + {}_0\text{n}^1 \rightarrow {}_{92}\text{U}^{236} \rightarrow {}_{56}\text{Ba}^{141} + {}_{36}\text{Kr}^{92} + 3 {}_0\text{n}^1 + Q(\text{energy})$	Example: ${}_1\text{H}^2 + {}_1\text{H}^2 = {}_2\text{He}^4 + E(\text{rnergy})$

13. Define nuclear fission with example (Jan 2016, 2014). (Or) What are nuclear reactions? Give some explain? (May2018, Sep2020, Nov 2017)

Nuclear fission is a nuclear reaction in which heavy nuclei after capturing a neutron splits up into two higher nuclei or nearly equal masses.

Here heavy nucleus uranium (z-92) splits up into two higher nucleus (Barium (z=56), Krypton (z=36)) after capturing a neutron.

14. What is chain reaction? (Jan 2016)

Chain reaction is a process in which the nuclear fission of an atom induces nuclear fission in another atom which again induces fission in another and so on.

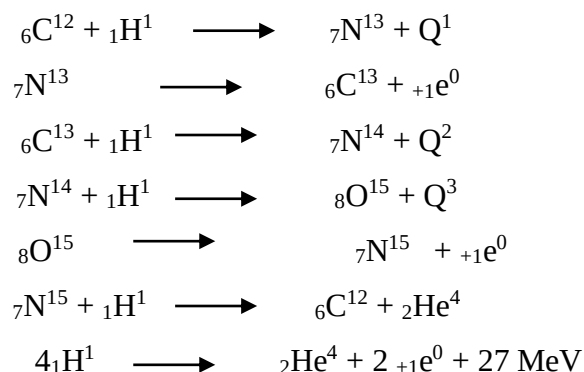
It is self-programming process in which number of neutrons goes on multiplying rapidly almost in geometrical progression during fission till whole of fissionable material is disintegrated.

15. What is thermo-nuclear reaction?

Thermo-nuclear reaction is a type of nuclear fusion reaction takes place at very high temp (10^9 °C) in which two lighter nuclei fuse together to form a heavier and stable nucleus with the release of large amount of energy.

16.What is carbon –nitrogen cycle? (May 2017)

Carbon nitrogen cycle is a cycle of nuclear reactions resulting into the conversion of hydrogen nuclei into helium nucleus carbon acting as a catalyst with the liberation of a large amount of energy per cycle .energy liberation in sun and other stars is supported to take place via carbon –nitrogen cycle.



17. What is breed reactor? (May 2014)

Breed reactor is a nuclear reactor which is used to convert fertile material like ${}_{92}\text{U}^{235}$ in to fissionable material.

18. What is moderator? (Jan2015)

Moderator is a substance which is used in a nuclear reactor to slow down the fast neutrons in the process of fission to thermal neutrons.

Commonly used moderator

1. High boiling point
2. Large scattering cross section
3. Small absorption cross section
4. Low atomic number.

Materials used:

1. Heavy water 2. Water
3. Graphite 4.beryllium oxide.

19. What is pressurized water reactor (PWR)?(Or) Define PWR materials used in nuclear reactors? (Nov/Dec 2018)

Pressurized water reactor (PWR) is a nuclear reactor in which water used as a moderator, coolant and reflector coolant and reflector is kept under pressure to prevent it from boiling at the operating temperature.

20.What is Boiling water reactor (BWR)?(Or) Define BWR materials used in nuclear reactors? (Nov/Dec 2018)

Boiling water reactor (BWR) is a nuclear reactor in which water is used as both coolant and moderator hence water is allowed to boil inside the reactor core itself and it is converted into steam.

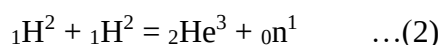
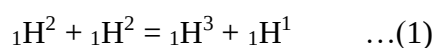
21. Define atomic mass unit? (May 2015, Nov 2018) (or) What is meant by atomic mass unit?

Atomic mass unit defined as accurately $1/12$ the mass of a carbon -12 atom

22. What is the principle used in D-D nuclear reactor?(Jan 2014) (Or) Give the principle of nuclear fusion reactors? (Nov 2019)

Nuclear fusion is the principle of fusion reactor. To produce energy from hydrogen, the hydrogen must be raised to very high temperature (10 to 100 million degrees). Under this condition, the electrons will be stripped away from the nuclei, then there will be only electrically charged nuclei and free electrons. This is called **plasma**. Nuclei and electrons will be moving very rapidly in random directions.

If the plasma is hot enough, nuclei overcome the electrical barrier, undergo following reactions (1) and (2) and produce more energy, increasing the plasma's temperature further. This heat would then be used as in an ordinary power plant, to generate electricity.



The hot plasma cannot be kept in a container since it has high temperature.

23. Calculated the radius of helium nucleus? (Nov 2019)

$$R \propto A^{1/3}$$

$$R = r_0 A^{1/3} \quad \dots(1)$$

where r_0 is the constant of proportionality and is equal to 1.3F

$$(1 \text{ Fermi, } F = 10^{-15} \text{ m}) R \propto A^{1/3}$$

$$R = r_0 A^{1/3} \quad \dots(1)$$

$$R = 1.3 \times 10^{-15} \text{ m} \times (4)^{1/3} = 52.92 \text{ F}$$

Pondicherry University (Part – B Questions)

1. Explain the construction and working of a Pressurized Heavy Water Reactor and boiling water reactor. What are the advantages for PWR and BWR?(Jan 2014 Jan 2013, 2016), (May2011,2012,2013, 2017,Nov,2016,2018) (OR) Discuss with neat sketch the construction and working of Boiling Water Reactor (BWR)

PRESSURIZED WATER REACTOR (PWR)

Definition: It is a nuclear reactor in which water is used as moderator coolant and reflector and is kept under pressure to prevent it from boiling at the operating temperature.

Principle:

Huge amount of heat energy is released when a heavy nucleus undergoes a nuclear chain fission process. This heat energy is converted into steam which is used for the production of electricity.

Description:

The nuclear power plant using a pressurized water reactor is shown in Fig 5.7.

PWR uses two circuits.

1. A primary circuit containing the reactor, pressuriser , one side of a heat exchanger and coolant pump.

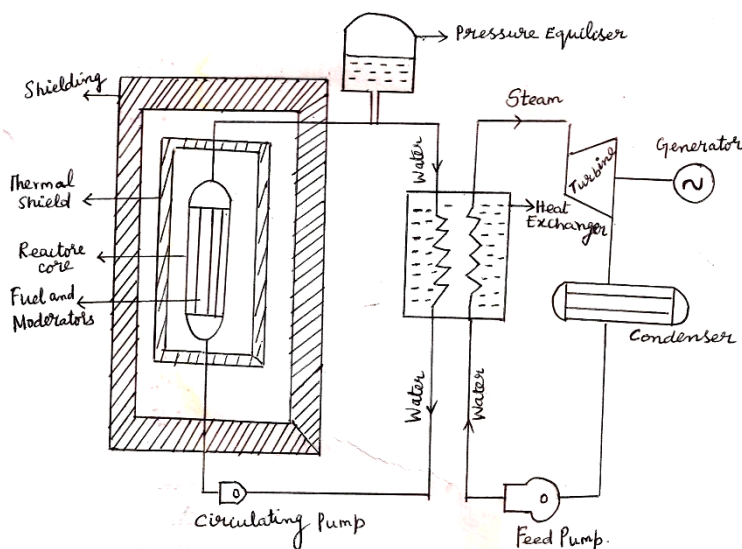


Fig 5.7.

2. A secondary circuit containing turbine, condenser, feed pump and the steam side of the heat exchanger.

Fuel

It uses enriched Uranium (contains more fissionable U^{235}) as fuel.

Moderator, reflector and coolant

Water (light or heavy) is used as moderator, reflector and coolant.

Pressure Equalizer

To maintain the constant pressure in the primary circuit throughout the load range, the pressurizing tank or pressurizer is included in the circuit. It is simply a pressure vessel with a heater at the bottom and a water spray at the top of the vessel and is filled with steam which maintains the primary pressure circuit.

If the primary circuit pressure drops the heater is operated which generates steam and increases the pressure of the primary circuit. If the primary circuit pressure becomes too high, cold water is sprayed into the steam in the pressurizer. This condenses the steam and reduces the primary circuit pressure.

Heat exchanger or steam generator

In the heat exchanger heat is carried by the coolant and is converted into steam.

Turbine

It converts heat energy into mechanical energy.

Electric generator

Here, mechanical energy of the turbine is converted into electrical energy.

Condenser

It is used to condense the steam by reducing the pressure of the expanded steam.

Circulating pump

It is used to circulate the water (coolant)

Feed pump

It is used to pump the condensed steam from the condenser to reactor core.

Operation

The water (coolant) in the primary circuit is pumped to the reactor core. The coolant is heated by the heat energy released due to nuclear fission in the reactor core. The hot water passes through the heat exchanger. In the heat exchanger, the coolant transfer the heat energy to feed water and steam is generated. The steam turbine is connected to the generator for the production of electrical power. The water from the heat exchanger is again circulated by the coolant pump.

To prevent boiling of water in the reactor it is kept under a pressure of 14 MN/m^2 . This enables water to carry more heat from the reactor. The pressuriser regulates the pressure in the circuit to safe guard the reactor from the power fluctuation.

Advantage:

1. Water which is cheaply available in plenty is used for coolant, moderator and reflector.
2. It is compact in size and has high power density (65 KW/litre)
3. Number of control rods required is less.
4. Steam is not contaminated by radiation. Hence maintenance of turbine is normal.
5. It gives complete freedom to inspect and maintain the turbine, feed heaters and condenser during operation.

Disadvantage:

- (i) Since there is high primary circuit pressure, it requires strong pressure vessel. Hence it needs high capital cost.
- (ii) Thermal efficiency of the plant is low since low pressure is maintained in the secondary circuit
- (iii) Corrosion problem occurs at high pressure and high temperature of water. Hence proper vessel is necessary to prevent corrosion. It further increases the cost of the plant.
- (iv) Reprocessing of fuel is difficult since it is affected by radiation.
- (v) For changing the fuel, it is necessary to shutdown the reactor. It requires more than one or two months.

BOILING WATER REACTOR (BWR)

Definition:

It is a nuclear reactor in which water in contact with fuel elements is used as coolant, moderator and reflector.

Principle:

Huge amount of heat energy is released when a heavy nucleus undergoes a nuclear fission chain reaction. This heat is converted into steam which is used in the production of electricity.

Description:

The nuclear plant using boiling water reactor is shown in Fig 5.8. It consists of reactor, turbine, condenser and feed pump.

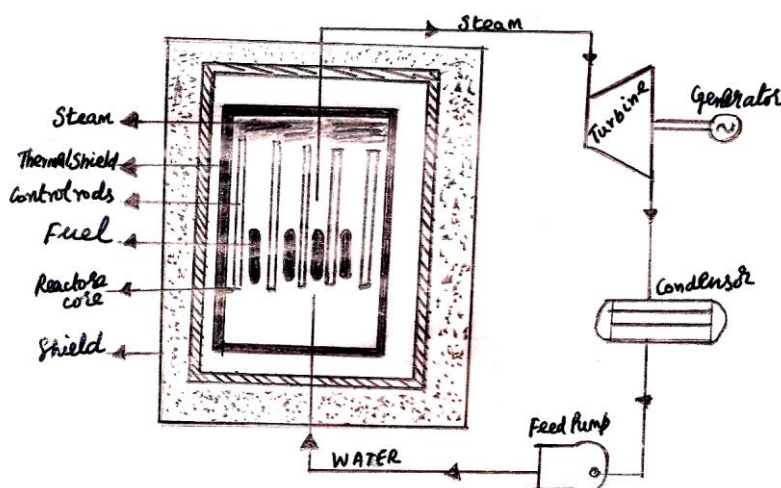


Fig 5.8. Boiling Water Reactor

Reactor:

Nuclear fission chain reaction takes place and huge amount of heat energy is released in the reactor.

Fuel:

It uses enriched Uranium as fuel.

Coolant, Moderator & reflector:

Water is used as coolant, moderator and reflector as in the PWR.

Turbine:

It converts heat energy into mechanical energy.

Electric generator:

Mechanical energy is converted into electrical energy.

Condenser:

It is used to condense the steam by reducing the pressure of the expanded steam.

Feed Pump:

It is used to pump the condensed steam from the condenser to the reactor core.

Working:

Water (coolant) is pumped into the reactor at the bottom. This water is heated by the heat released due to the nuclear fission in the reactor core and gets converted into steam. The steam leaves from the top of the reactor to the turbine which is connected to the generator for the production of electrical power. Exhaust steam from the turbine passes through the condenser and is condensed. The condensed steam is re-circulated again using feed pump.

Advantages:

1. As water is allowed to boil inside the reactor, the pressure inside the reactor is small. Hence high pressure vessels are not required. It reduces the cost considerably.
2. Steam generator, pressuriser, circulating pump and connecting piping are not required. Hence the cost is further reduced.
3. Thermal efficiency of BWR plant is more than PWR.
4. It is a self controlled reactor as the reactivity automatically is reduced if the vapor is not dense enough to moderate the neutrons effectively.

Disadvantages:

1. It cannot meet a sudden increase in power demand.
2. The steam leaving the turbine is slightly radioactive. Hence shielding of turbine and piping is essential.
3. The size of the vessel is large since it has low power density. There is a possibility of burn out of fuel in the reactor as boiling of water on the surface of fuel is allowed

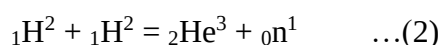
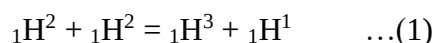
2. Discuss in detail about the nuclear fusion reactors .(OR) Give the basic principle of nuclear fusion reactors with suitable example? (Jan 2014,May 2015,Nov 2017)

Nuclear Fusion Reactor

Principle of Fusion reactor

Nuclear fusion is the principle of fusion reactor. To produce energy from hydrogen, the hydrogen must be raised to very high temperature (10 to 100 million degrees). Under this condition, the electrons will be stripped away from the nuclei, then there will be only electrically charged nuclei and free electrons. This is called **plasma**. Nuclei and electrons will be moving very rapidly in random directions.

If the plasma is hot enough, nuclei overcome the electrical barrier, undergo following reactions (1) and (2) and produce more energy, increasing the plasma's temperature further. This heat would then be used as in an ordinary power plant, to generate electricity.



The hot plasma cannot be kept in a container since it has high temperature. There are two approaches to confining the plasma

1. Magnetic confinement
2. Inertial confinement

1. **Magnetic confinement:** The trajectories of fast-moving electrically charged particles are bent in a magnetic field. Plasma physicists have devised very complex arrangements of magnets, so that the electrons and ions are kept within a finite volume, when a particle approaches the edge of the volume, it feels a magnetic force that turns it back into the volume.

2. **Inertial confinement:** In inertial confinement, a small pellet of solid hydrogen fuel is hit on all sides by many laser beams. This compresses the pellet and heats it to a fusion temperature. The pellet is quickly vaporized and begins to dissipate, but it may stay together long enough so that fusion can generate additional heat. The object in each of these techniques is not to keep the plasma confined forever, but to keep it confined long enough so that fusion can generate more heat energy than was used to trigger the fusion.

Construction and Working of fusion reactor

a) The fusion reactor heats fuel (Deuterium and Tritium) to form high-temperature plasma. It squeezes the plasma so that fusion can take place. The power needed to start the fusion reaction is about **70 Megawatts**, but the output power is **500 Megawatts**. The fusion reaction lasts from **300 to 500 seconds**. (Eventually, there will be a sustained fusion reaction).

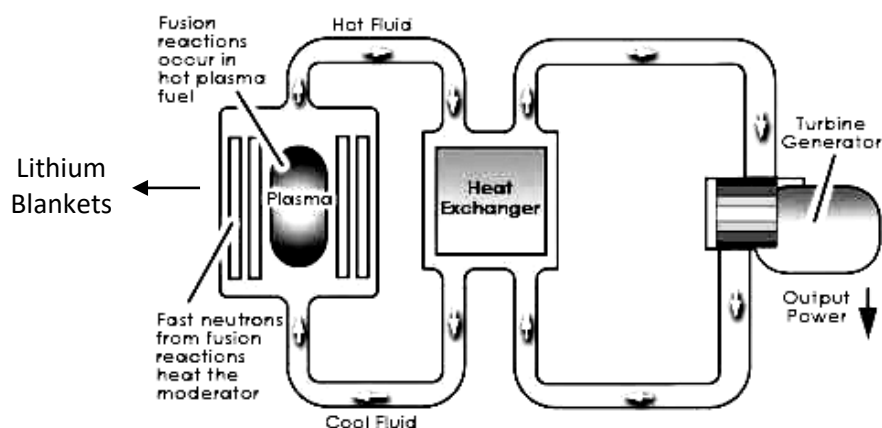


Fig.5.5. Fusion reactor

- b) The lithium blankets outside the plasma reaction chamber absorb high-energy neutrons from the fusion reaction to make more Tritium fuel. The blankets also get heated by the neutrons as shown in Fig. 5.5.
- c) The heat is transferred by a water-cooling loop to a heat exchanger to make steam.
- d) The steam drives the electrical turbines to produce electricity.
- e) The steam condenses back into water to absorb more heat from the reactor in the heat exchanger.

3.Explain the general properties of Nucleus on the basis of size, mass and density. (Jan 2014,Jan 2013,2016), (May 2013,2017)

(i). Nuclear size

If the nucleus is assumed to be spherical, an empirical relation is found to hold good between the radius of the nucleus R and its mass number A . It is given by

$$R \propto A^{1/3}$$

$$R = r_0 A^{1/3} \quad \dots(1)$$

where r_0 is the constant of proportionality and is equal to $1.3F$

(1 Fermi, $F = 10^{-15} \text{ m}$)

(ii). Nuclear density

The nuclear density ρ_N can be calculated from the mass and size of the nucleus.

$$\rho_N = \frac{\text{Nuclear mass}}{\text{Nuclear volume}}$$

Nuclear mass = $A m_N$ where, A = mass number and m_N = mass of one nucleon and is approximately equal to $1.67 \times 10^{-27} \text{ kg}$.

$$\begin{aligned} \text{Nuclear volume} &= \frac{4}{3} \pi R^3 \\ &= \frac{4}{3} \pi (r_0 A^{1/3})^3 \\ &= \frac{4}{3} \pi (r_0^3 A) \\ &= \frac{4}{3} \pi r_0^3 A \end{aligned} \quad \dots(1)$$

$$\therefore \rho_N = \frac{A m_N}{\frac{4}{3} \pi r_0^3 A}$$

Substituting the known values, the nuclear density is calculated as $1.816 \times 10^{17} \text{ kg m}^{-3}$ which is almost a constant for all the nuclei irrespective of its size.

(iii). Nuclear charge

The charge of a nucleus is due to the protons present in it. Each proton has a positive charge equal to $1.6 \times 10^{-19} \text{ C}$.

$\therefore$ The nuclear charge = Ze , where Z is the atomic number.

(iv). Nuclear mass

As the nucleus contains protons and neutrons, the mass of the nucleus is assumed to be the mass of its constituents. Assumed nuclear mass = $Zm_p + Nm_n$, where m_p and m_n are the mass of a proton and a neutron respectively. However, from the measurement of mass by mass spectrometers, it is found that the mass of a stable nucleus (m) is less than the total mass of the nucleons. i.e., mass of a nucleus, $m < (Zm_p + Nm_n)$, $Zm_p + Nm_n - m = \Delta m$, where Δm is the mass defect.

4. (a). Explain the classification of nuclear reactors. (May2011,2012,2017)

Classification of nuclear reactors:

Nuclear reactor may be classified under different ways they can be classified as

- (a). Thermal reactors in which fission is induced primarily by thermal neutrons.
- (b). Intermediate reactors in which fission is induced by epithermal neutrons.
- (c). Fast neutron reactors in which fission is induced by fast neutrons.

Nuclear reactors can also be classified as

1. Homogeneous reactors:

The nuclear fuel and the moderator represent a uniform mixture.

2. Heterogeneous reactor:

Separate fuel rods are inserted in the moderator. Nuclear reactors may also be classified according to the type of the moderator, such as

- (i) Graphite reactor
- (ii). Heavy water reactor

3. Breeder reactor:

A reactor which is used to convert fertile material like ${}_{92}\text{U}^{235}$ into fissionable material is called the breeder reactor.

(b) Explain the construction and working of a thermal reactor. (May2011), (May 2012,2017) (OR) Discuss in detail the various materials used in nuclear reactors.(Jan 2016,may 2016) (OR) Describe the properties of material used in nuclear reactor? (Dec2019).

Nuclear reactor

A nuclear reactor is a device in which the nuclear fission reaction takes place in a self sustained and controlled manner. The first nuclear reactor was built in 1942 at Chicago USA.

Essential components of Reactor and its functions

Following are the essential components of a nuclear reactor.

- a) Reactor core
- b) Moderator
- c) Reflector
- d) Coolant
- e) Control system
- f) Shielding

a) Reactor core:

It is the heart of the reactor; here the nuclear fission takes place at a steady controlled rate. The fission energy is liberated in the form of heat.

The reactor core consists of fuel, moderator, control rods and the space for coolant. The reactor core is cylindrical in shape and is contained in a pressure vessel.

Reactor fuel: The commonly used fissionable materials are the Uranium isotopes, Thorium isotopes and the Plutonium isotopes. It is generally taken in the form of rods.

b) Moderator: It is a material, which reduces the speed of fast neutrons in the process of nuclear fission to thermal neutrons. The commonly used moderators are Heavy water, Ordinary water, Graphite and Beryllium oxide. The purpose of moderators is to reduce the energy of neutrons, but it should not absorb neutrons.

c) Reflector: It is a material, which is used to reflect back some of the escaping neutrons into the reactor core. It will increase the neutron flux in the interior of the core. The materials used for reflector is the same as the materials used for the moderators.

d) Coolant: The purpose of coolant is to remove heat produced inside the reactor core. The coolant is taken generally in the form of liquid or gas. It is circulated through the reactor core by means of a coolant pump. The following materials are used as coolants:

1. Water(light water or heavy water)
2. Gases (helium, carbon dioxide)
3. Liquid metals (sodium and organic liquid).

e) Control systems (Control Rods):

The control system is necessary in a nuclear reactor for the following purposes

1. To control the rate of fission.
2. To maintain the chain reaction at a steady state.
3. To shut down the reactor at the steady state.

The commonly used control rods are made up of elements like Boron or Cadmium. The control rods are inserted into the core and they pass through the space in between the fuel tubes and through the moderator. By pushing them in or pulling out, the reaction rate can be controlled.

f) Shielding

In a nuclear reactor, a large amount of penetrating radiations like α -rays, β -rays, in addition to neutrons are emitted. These radiations are very harmful to the technicians working around the reactor. Hence a thick shield, in the form of concrete walls of many meters thick, always surrounds a reactor

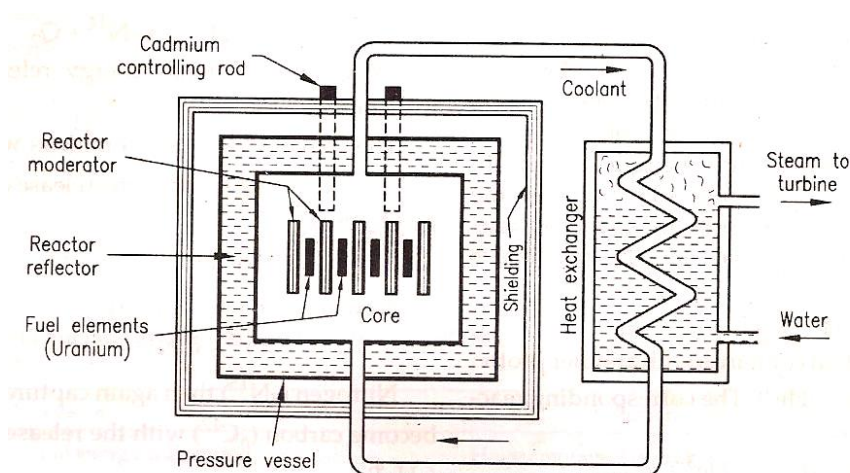


Fig.5.6. Nuclear reactor

Working of the reactor

Neutrons produced by action of alpha (α) particles on Polonium or Beryllium is slowed down and is used initially to cause fission of say U^{235} nuclei. Thus chain reaction sets in. The heat energy is produced during nuclear fission reaction. This heat energy is absorbed by the coolant and is used to drive the machinery producing electricity.

Power output of a nuclear reactor

The fission rate of a reactor i.e., total number of nuclei undergoing fission per second in a reactor is $= nv\sigma NV = \phi\sigma NV$

Where, n = average neutron density i.e., number per m^3 ,

v = average speed in ms^{-1} ,

$\phi = nv$ = average neutron flux,

N = number of fissile nuclei per m^3 ,

σ = fission cross section in m^2 ,

V = volume of the nuclear fuel.

Thermal reactor:

In these reactors, the fast neutrons are slowed with the use of moderators. The slow neutrons are absorbed by the fissionable fuel and the chain reaction is

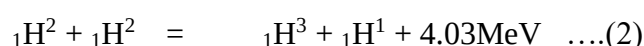
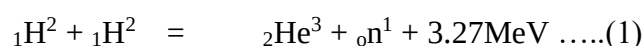
maintained. The moderator is most essential component in these reactors. (The construction and working of these reactors are same as explained earlier).

5. Explain how D-D and D-T reactions take place in fusion. (OR) Discuss the D-D and D-T nuclear fusion reaction with equation(OR) Explain nuclear reaction in earth and stars (Jan 2016,Nov 2017,2019,May2016)

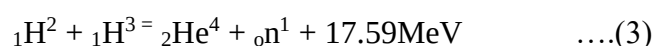
Deuterium-Tritium cycle of fusion

When two deuterons come close together they may just bounce off each other. The four fusion reactions which can occur with Deuterium and Tritium can be considered to form a Deuterium-Tritium cycle. The four reactions:

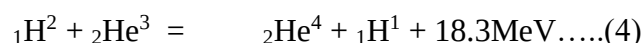
(Deuterium – deuterium fusion)



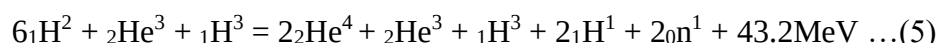
(Deuterium – Tritium fusion)



(Deuterium – Helium-3 fusion)

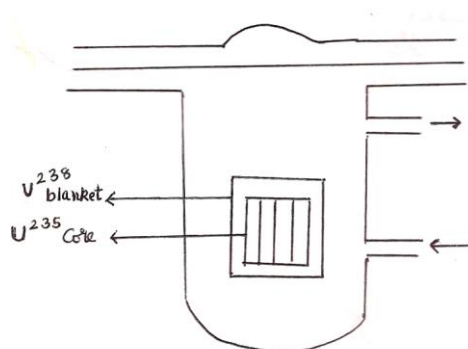


The above equations can be combined as



6.Explain the construction and working of FBTR. (2014Jan,Nov2017,Dec2015)

The below figure 5.9 shows a fast breeder reactor system. In this reactor the core containing U^{235} surrounded by a blanket (a layer of fertile material outside the core) like U^{238} . In this reactor no moderator is used. The fast moving neutrons liberated due to fission U^{235} are absorbed by U^{238} which gets converted into fissionable material Pu^{239} which is capable of sustaining chain reaction.

**Fig 5.9**

Thus this reactor is important because it breeds fissionable material from fertile material U^{238} available in large quantities. Fast breeder reactors are better than conventional reactor from the point of view of safety and thermal efficiency.

For India which already is fast advancing towards self reliance in the field of nuclear power technology the fast breeder reactor becomes important in view of massive reserves of Thorium and the finite limits of its Uranium resources. The research and development efforts in the fast breeder reactor technology will have to be stepped up considerably if nuclear power generation is to make any impact on the countries total energy needs in the distant future.

LIQUID METAL FAST BREEDER REACTOR

Characteristics:

- Fuel : Mixed Fuel (PUO and UO)
- Coolant : Liquid Sodium
- Moderator : No moderator is used
- Condenser : It is used to condense the steam by reducing the pressure of the steam.
- Generator : Mechanical energy of the turbine is converted into Electrical Energy.
- Power output: 40 MW thermal and 13.2MW electrical.

Fast breeder reactors are designed to create or breed new fissile material for producing useful electrical power. Most produce fissile Plutonium from U^{238} . The fuel rods in the core region contain a mixture of fissile Pu^{239} and U^{238} . The active core region is surrounded by a blanket of fertile U^{238} . This blanket region captures neutrons that would otherwise be lost through leakage, it produces additional fissile material. A fast neutron reaction with U^{238} producing Pu^{239} is shown below.

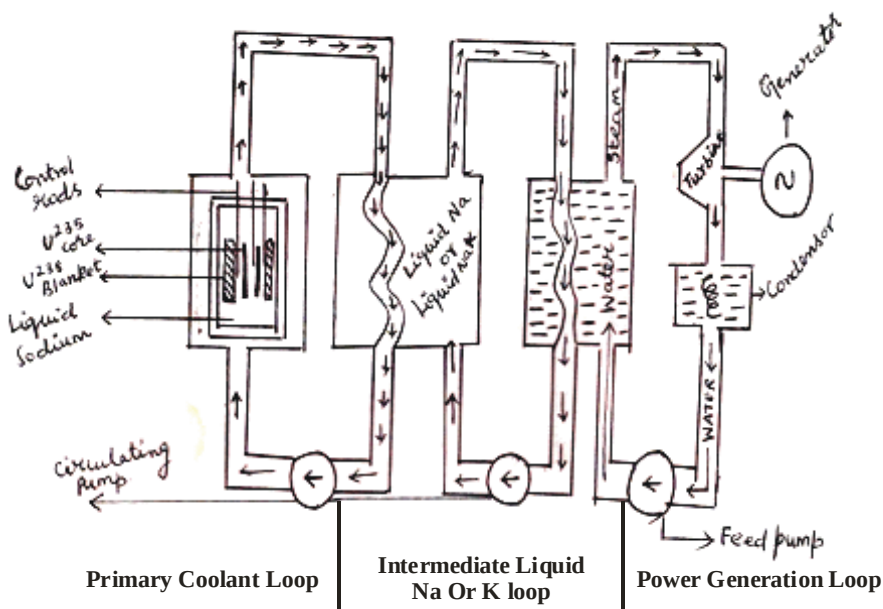
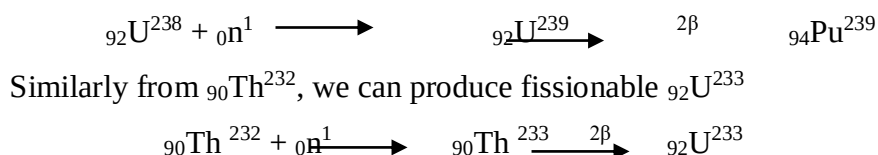


Fig. 5.10 FBTR Reactor

By suitable controlled conditions more ${}_{94}\text{Pu}^{239}$ can be produced in the natural Uranium blanket than U^{235} that is used up in the reaction. For this reason this type of reactor is called fast breeder reactor. Figure 5.10 shows the parts of FBTR reactor.

In October 1985, the Indira Gandhi centre for Atomic Research (IGCAR) and Bhabha Atomic Research centre (BARC) jointly designed, constructed and operated the reactor. The FBTR is liquid metal fast breeder reactor. The reactor uses Plutonium- Uranium mixed Carbide fuel and liquid sodium as coolant.

The fuel is an indigenous mixture of 70% Plutonium Carbide and 30% Uranium Carbide. The reactor was designed to produce 40MW thermal power and 13 MW electrical powers. During the operation of the reactor heat is produced and this heat is minimized using proper coolant. A coolant with excellent heat transfer properties is required to minimize the temperature drop from the fuel surface to the coolant and also it must be non-moderating, considering the above reasons water is ruled out.

The coolants for fast breeder reactor are

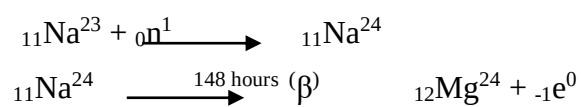
1. Liquid Metal (Na or NaK)
2. Helium (He)
3. Carbon-di-oxide

The best coolant for fast breeder reactor is sodium in liquid because of the following reasons.

1. It has low absorption cross-sectional area.
2. It possess good heat transfer properties and low pressure property.
3. It does not react to any of the structural material in primary circuits.
4. The boiling point of sodium at atmospheric pressure is very high, so it can use unpressurized in a reactor.

The heat produced in the core is removed by two hydraulically coupled primary sodium loop which in turn transfer the heat to two independent secondary loops containing Sodium Potassium alloy. The secondary coolant while flowing through the intermediate heat exchanger transfers its heat to feed water. Proper heat insulation is provided on all pipelines to maintain sodium in liquid form.

Liquid sodium is more effective as a medium for heat transfer, but it a more reactive and undergoes radioactive decay by capturing a neutron.



Because of the induced radioactivity of liquid Sodium an intermediate loop also using Na or NaK as coolant is between the primary loops and will act as a radioactive coolant.

Advantages:

1. The high concentration of fissile fuel and the absence of moderator make core of a fast reactor smaller than the thermal reactor of same power.

Merits of the Fast breeder reactors:

1. Extraction of 100 times more energy from Uranium than via light water reactors, enabling the use of depleted Uranium which is regarded as an unstable waste at present.
2. Much higher power generation than PHWR:
 - a. PHWR : 73 tons $\rightarrow$ 15 MW
 - b. FBR : 73 tons $\rightarrow$ 350 MW
3. Lower pollution factor
4. Much higher Thermodynamic efficiency (40% vs. 30%)

7.Explain the nuclear fusion reaction which takes place in the sun and stars.

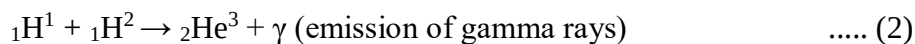
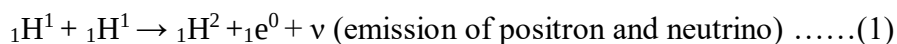
(OR) Explain in detail the P-P and C-N cycle of nuclear fusion.

(May 2018,Nov2017,Jan2015,2014,May2014,Sep2020)

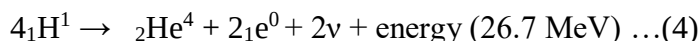
Fusion is the source of stellar energy. It has been estimated that the total energy radiated by sun is about 3.8×10^{26} joule per second. The origin of such a tremendous amount of energy is the fusion of protons into helium. Proton–proton cycle and carbon–nitrogen cycle are the two important types in which nuclear fusion takes place in sun and stars.

Proton-Proton cycle

The production of stellar energy is by thermo nuclear reactions in which protons are continuously transformed into helium nuclei. The proton-proton cycle takes place in **sun**.



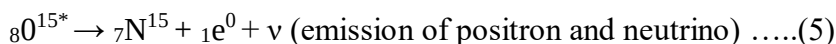
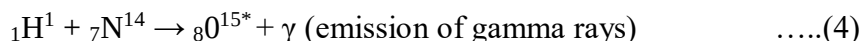
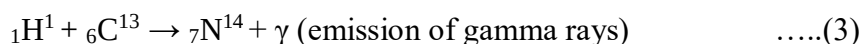
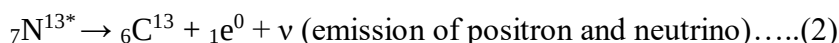
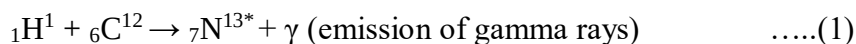
The reaction cycle is written as



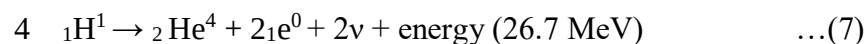
Thus four protons fuse together to form an alpha particle and two positrons with a release of large amount of energy.

Carbon-Nitrogen cycle

The carbon-nitrogen cycle takes place in **stars**. The following cycle of reactions take place in carbon – nitrogen cycle in which carbon acts as a catalyst.



The overall reaction of the above cycle is given as



8. Write a short note on disintegration in fission? (Dec 2018, May 2018,2014,Jan 2015)

Nuclear fission

The process of breaking up of the nucleus of a heavier atom into two fragments with the release of large amount of energy is called **nuclear fission**. The fission is accompanied by the release of neutrons. The fission reactions with ${}_{92}\text{U}^{235}$ are represented as

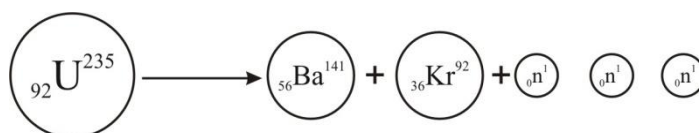
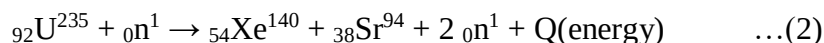


Fig. 5.2 Fission in Uranium

In the above examples the fission reaction is taking place with the release of 3 neutrons and 2 neutrons.

9.What is a nuclear fission reaction? How is the energy released in a nuclear fission reaction? What are the conditions needed to produce a continuous nuclear fission reactions? (Nov 2018,Nov 2017) (OR) Explain terms nuclear fission and fusion what is the source of energy released in this reactions?

Nuclear fission

The process of breaking up of the nucleus of a heavier atom into two fragments with the release of large amount of energy is called **nuclear fission**. The fission is accompanied by the release of neutrons. The fission reactions with ${}_{92}\text{U}^{235}$ are represented as

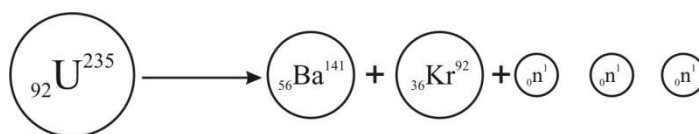
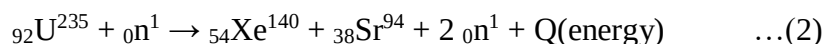


Fig. 5.2 Fission in Uranium

In the above examples the fission reaction is taking place with the release of 3 neutrons and 2 neutrons.

Energy released in fission

Let us calculate the amount of energy released during the fission of ${}_{92}\text{U}^{235}$ with a neutron. The fission reaction is



$$\text{Mass of } {}_{92}\text{U}^{235} = 235.045733 \text{ amu, Mass of } {}_0\text{n}^1 = 1.008665 \text{ amu}$$

$$\text{Total mass of the reactant} = 236.054398 \text{ amu}$$

$$\text{Mass of } {}_{56}\text{Ba}^{141} = 140.9177 \text{ amu}$$

$$\text{Mass of } {}_{36}\text{Kr}^{92} = 91.8854 \text{ amu}$$

$$\text{Mass of 3 neutrons} = (3 \times 1.008665) = 3.025995 \text{ amu}$$

$$\text{Total mass of the products} = 235.829095 \text{ amu}$$

$$\therefore \text{Mass defect} = 236.054398 - 235.829095 = 0.225303 \text{ amu}$$

$$\text{As, } 1 \text{ amu} = 931 \text{ MeV, energy released in a fission}$$

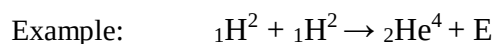
$$= 0.225303 \times 931 \text{ MeV} = 200 \text{ MeV}$$

Condition to be satisfied for sustained chain reaction:

1. The proportion of U^{235} to U^{238} increase the probability of fission capture in neutrons.
2. The chain reaction can be sustained to possible only if multiplication.
 $K = 1$ critical state chain reaction; $K > 1$ super critical chain reaction.
3. There should be a moderator to slow down the neutrons to thermal energies without absorbing them.
4. Material used as moderators and uranium sample should be over 99.99% pure.
5. There is a certain size called the critical size for the system.

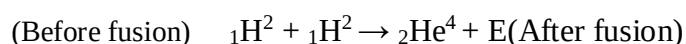
Nuclear fusion

It is the process in which two lighter nuclei are combined to fused together to form a heavier and stable nucleus.



Energy released in fusion

Let us calculate the amount of energy released during the fusion



Before fusion

$$\text{Mass of single deuterium} = 2.01478 \text{ amu}$$

$$\text{But mass of 2 deuterium atoms} = 2 \times 2.01478 = 4.02956 \text{ amu}$$

After fusion

$$\text{Mass of helium atom} = 4.00388 \text{ amu}$$

$$\text{Mass defect} = 4.02956 - 4.00388 \text{ amu}$$

$$= 0.02568 \text{ amu}$$

$$1 \text{ amu} = 931 \text{ MeV}$$

$$\text{Therefore the energy released} = 0.02568 \times 931 \text{ MeV}$$

$$= 23.91 \text{ MeV}$$

10. Draw a binding energy curve? What information do we get from such a curve?(Nov2017, Dec 2019) (OR) Explain binding energy diagram?

It is found that the binding energy per nucleon varies from element to element. A graph is plotted with the mass number of the nucleus along the X-axis and binding energy per nucleon along the Y-axis as shown in the figure 5.1.

(i) The binding energy per nucleon increases sharply with mass number A upto 20. It increases slowly after $A = 20$. For $A < 20$, there exists recurrence of peaks corresponding to those nuclei, whose mass numbers are multiples of four and they contain not only equal but also even number of protons and neutrons. Example: ${}^2\text{He}^4$, ${}^4\text{Be}^8$, ${}^6\text{C}^{12}$, ${}^8\text{O}^{16}$, and ${}^{10}\text{Ne}^{20}$. The curve becomes almost flat for mass number between 40 and 120. Beyond 120, it decreases slowly as A increases.

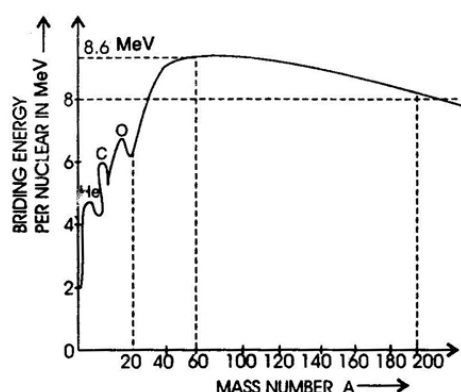


Fig.5.1. Variation of Binding per nucleon as a function of A

- (ii) The binding energy per nucleon reaches a maximum of 8.6 MeV at $A=56$, corresponding to the iron nucleus (${}_{26}\text{Fe}^{56}$). Hence, iron nucleus is the most stable.
- (iii) The average binding energy per nucleon is about 8.5 MeV for nuclei having mass number ranging between 40 and 120. These elements are comparatively more stable and non radioactive.
- (iv) For higher mass numbers the curve drops slowly and the $\frac{BE}{A}$ is about 7.6 MeV for Uranium. Hence, they are unstable and radioactive.
- (v) The lesser amount of binding energy for lighter and heavier nuclei explains nuclear fusion and fission respectively. A large amount of energy will be

liberated if lighter nuclei are fused to form heavier one (fusion) or if heavier nuclei are split into lighter ones (fission).